SMART INDIA HACKATHON 2025

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GyaanGuru

Problem Statement ID - SIH25199

Problem Statement Title -

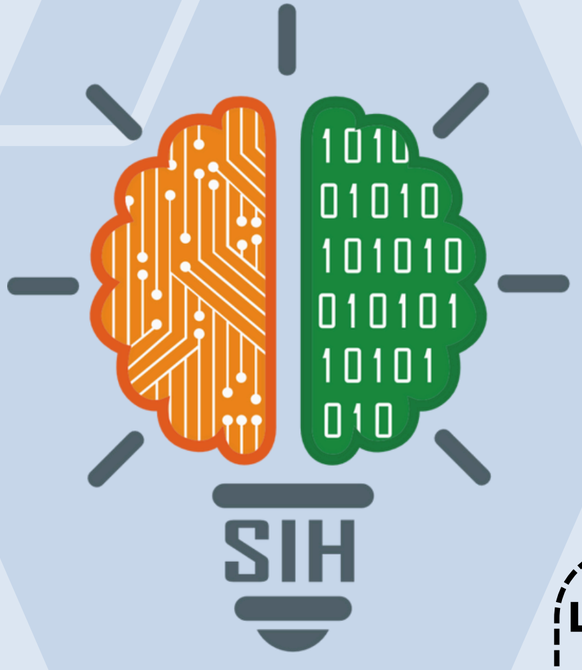
AI-Powered Personalized Learning Path Generator

Theme - Smart Education

PS Category - Software

Team ID - 65735

Team Name - Qbitron



LINKS (CLICK ME):

GitHub

YouTube

Qbitron

GyaanGuru

The Right Path to Your Future Skillset

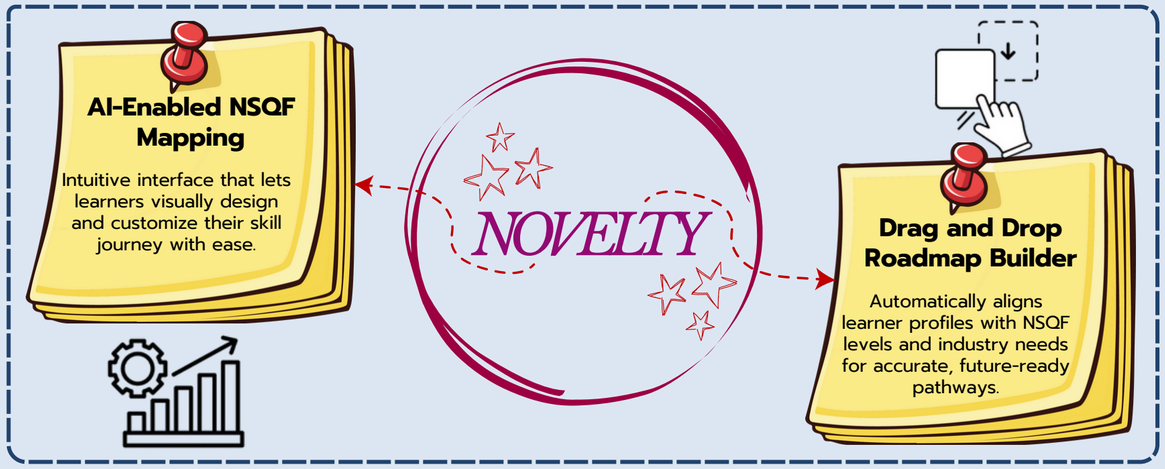
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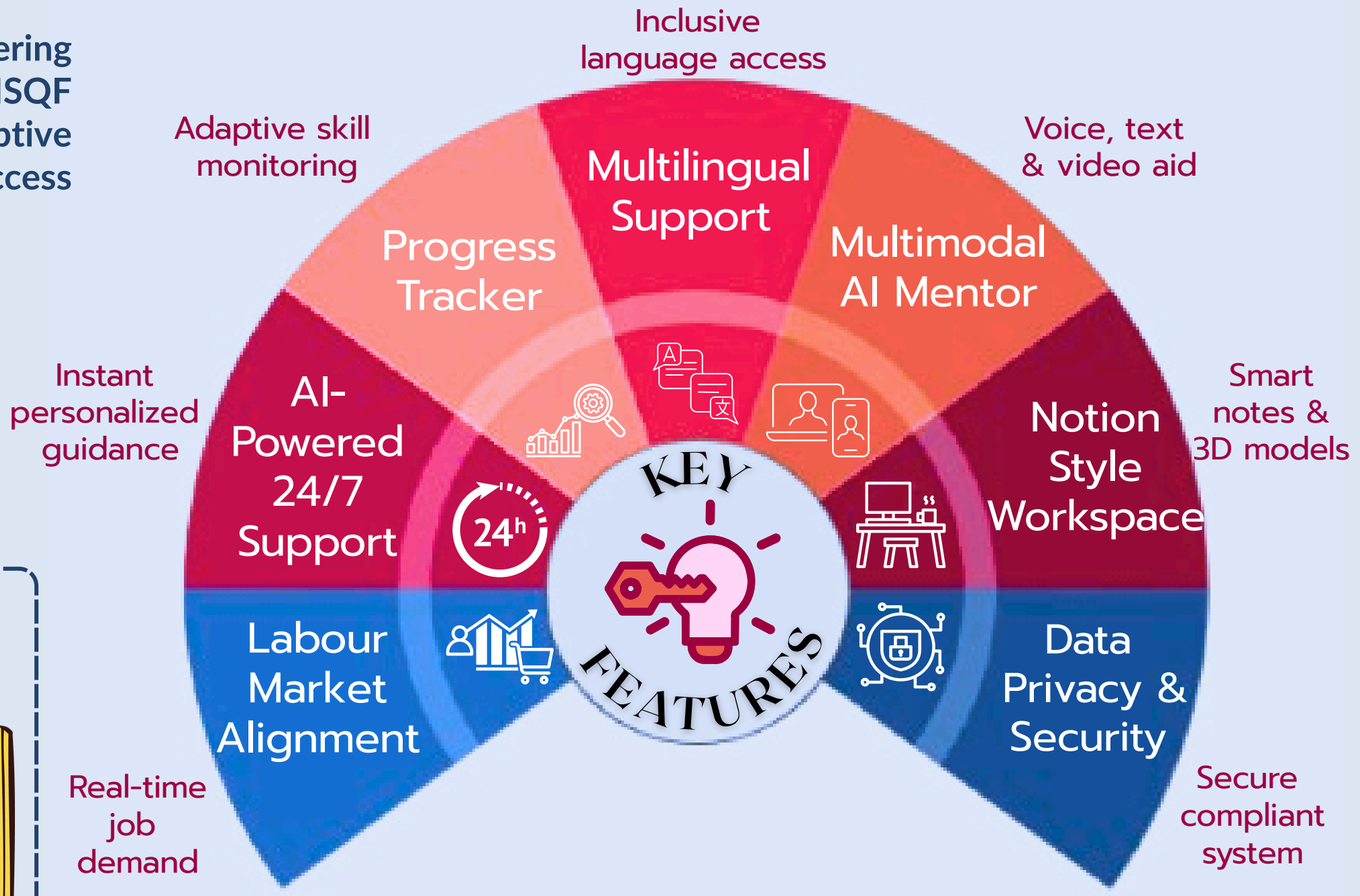
Overview

Our AI system designs personalized vocational pathways by considering each learner's background and goals, and aligning them with NSQF qualifications and current industry requirements. It provides adaptive courses, certifications, and apprenticeships, along with multilingual access and dashboards for learners, trainers, and policymakers.

Problem Addressed

- Bridges the gap between learners and industry requirements
- Reduces skill mismatches through tailored learning paths
- Provides personalized guidance for career growth
- Tracks learner progress and evolving skill demand
- Supports employability with future-ready training recommendations





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TECHNICAL APPROACH

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FRONTEND

NEXT.JS

Framework for building React-based web app

Handles:
- Pages, routing, server-side rendering

Used for:
- Learner dashboard
- Drag-and-drop roadmap UI
- Notion-style MD editor

BACKEND & AI ENGINE

EXPRESS JS

Node.js framework for REST APIs

Handles:
- Roadmap generation requests
- YouTube API integration
- Learner profile CRUD operations

CORE FUNCTIONALITY

FLASK

Python backend for AI-heavy tasks

Handles:
- Gemma API requests for roadmap / NSQF mapping
- Sarvam AI calls (multimodal AI mentor)
- Trellis Text-to-3D model generation

DATA & STORAGE

SUPABASE

Database + Auth + Storage

Stores:
- Learner profiles
- Roadmaps
- Notes / 3D models / videos
- Handles authentication (email/password, JWT)

PERSONALIZED AI CONTENT

sarvam.ai
- Provides fast, domain-specific reasoning in Indian contexts.
- Supports multilingual understanding and responses.

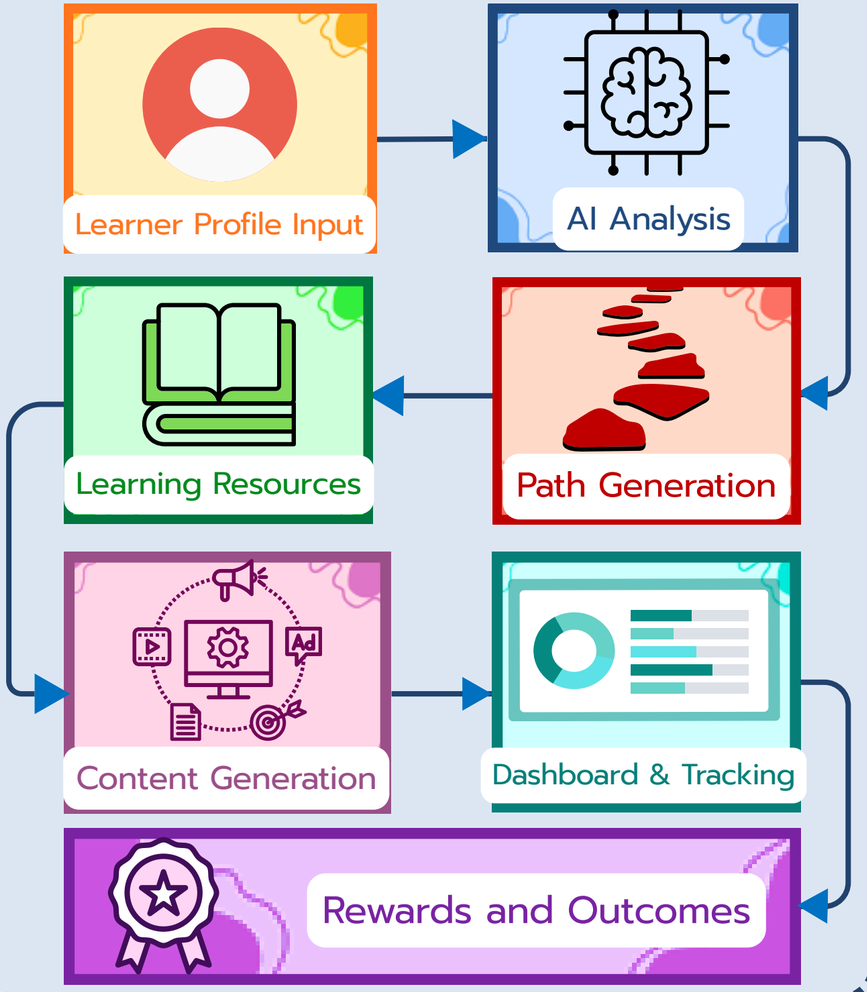
Gemma
- Multimodal AI mentor
- Reads screen content, answers questions, provides tutoring support

trellis
- Converts text prompts → 3D models for interactive learning
- Embed in Notion-style editor for hands-on exploration

<MANIM>
- AI video generation to dynamically generate videos for skillful learning

coursera Skill India API
- Pulls relevant learning videos for courses or weekly recommendations

PROCESS FLOW



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FEASIBILITY AND VIABILITY

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FEASIBILITY ANALYSIS

Inclusivity & Engagement
Inclusive & engaging via multilingual support, AI mentor, and interactive 3D/AI content.

Proven AI Capability
Proven AI models already available for learner profiling and personalized path generation.

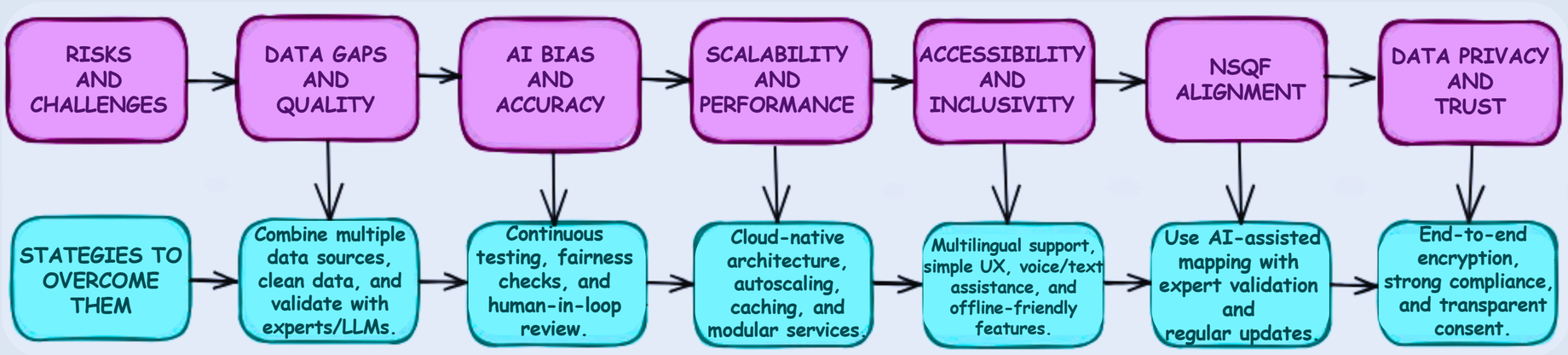
Data Availability
Reliable data access from NSQF, skill databases, and job market feeds for accurate mapping.

Scalable & Secure
Scalable & secure design with cloud architecture to support millions while keeping data safe.

Policy Alignment
Aligned with policy through NSQF-based pathways that support NCVET/MSDE initiatives.

Industry Relevance
Industry-linked using real-time labor market trends to guide learners toward in-demand skills.

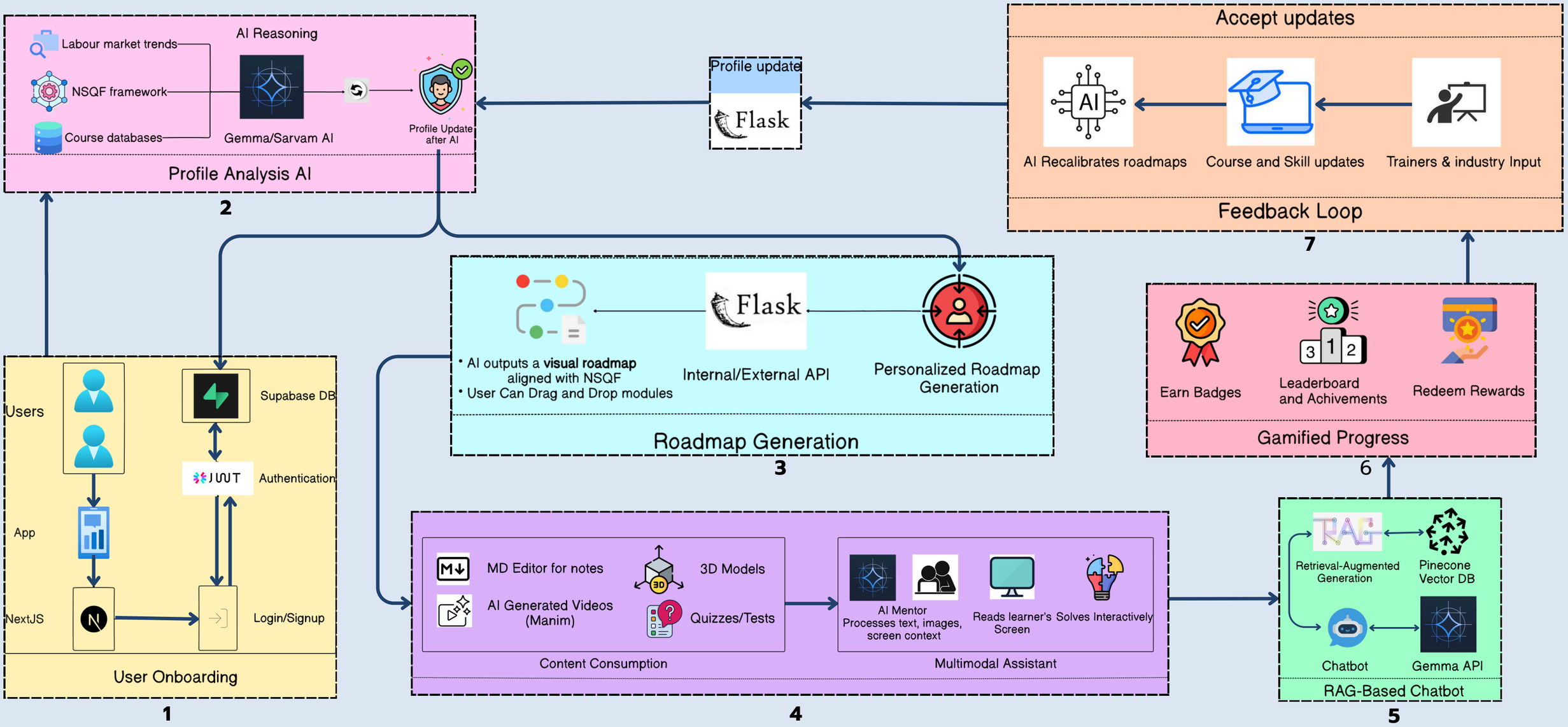
COMPETITORS	NSQF-MAPPING & COMPLIANCE	PERSONALIZED /ADAPTIVE ROADMAPS	MULTIMODAL MENTORING / AI SUPPORT	CONTENT VARIETY (VIDEOS, 3D, AI-GENERATED CONTENT, NOTES ETC.)	LABOR MARKET INTELLIGENCE /DEMAND ALIGNMENT	MULTILINGUAL /INCLUSIVE DESIGN
GyaanGuru	✓	✓	✓	✓	✓	✓
SKILLS Trainer	✓	✗	✗	✓	✗	✓
GUVI	✗	✗	✗	✓	✗	✓
swayam	✓	✗	✗	✓	✗	✓



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USE CASE DIAGRAM

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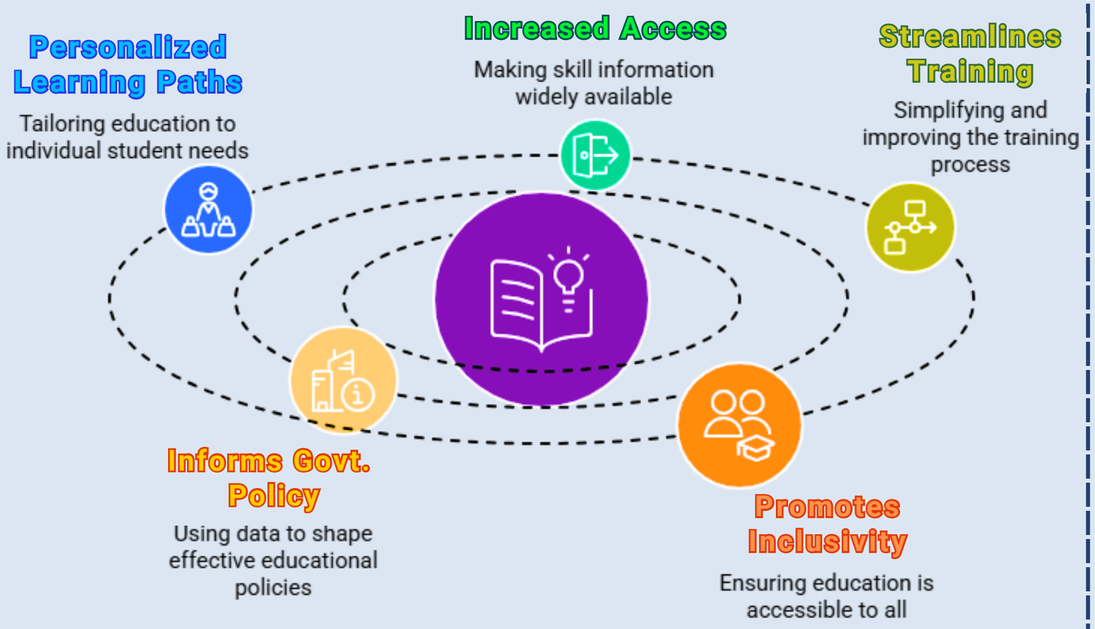


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IMPACT AND REFERENCES

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IMPACTS AND BENEFITS



...Future Scope...

Community-Based Add-ons: Learners connect, share resources, and learn collaboratively through peer-driven modules.

Self-Healing Correction Courses: Lets learners continue unfinished or difficult courses from other platforms on our system with adaptive guidance and personalized support.

RESEARCH AND REFERENCES

[1] Manim Community Documentation. [Online]. Available: https://docs.manim.community/en/stable/index.html?utm_source=chatgpt.com

[2] Microsoft, "Trellis: A System for Dynamic Neural Network Execution," arXiv preprint, arXiv:2412.01506, 2024. [Online]. Available: https://arxiv.org/pdf/2412.01506

[3] D. M. Córdova-Esparza, et al., "Analysis of AI-enhanced educational tools developed in India for linguistic minorities and disabled people," Education and Information Technologies, vol. 29, no. 3, pp. 1557–1576, 2024.

[4] A. K. Singh and M. Sharma, "Retrieval Augmented Generation (RAG)-based restaurant chatbot with AI testability," ResearchGate, 2024. [Online]. Available: https://www.researchgate.net/publication/381461839_Retrieval_Augmen_ted_Generation_RAG_based_Restaurant_Chatbot_with_AI_Testability

[5] P. S. Winkler and W. Söllner, "Interacting with educational chatbots: A systematic review," Education and Information Technologies, vol. 28, pp. 973–1018, 2023.